

## Session-2: Hands on ML/AI in Aquatic Remote Sensing

This session will include exploratory data analysis (EDA), training a model, evaluating the model with the metrics we discussed in the last session, and finally, inferencing the model on a satellite image to create a map. This exercise will be performed in Python.

It is recommended to have a laptop with at least 8GB of RAM, if you plan to run the exercise locally. You may otherwise use Google Colab, an online high performance Python environment as well – but comes with a different set of pre-requisites, which we will discuss *in situ* if anyone needs to use Google Colab. It is strongly recommended for everyone to create a GitHub profile, as that is from where I will primarily be distributing all the materials including data and starter code.

Please follow the checklist below, to make sure you are well prepared for the session tomorrow:

1. Download Anaconda from: <https://www.anaconda.com/download>
2. Clone/Download this repository from GitHub, from where all materials will be distributed:  
[https://github.com/Chintan2108/SCOR\\_Workshop\\_2025](https://github.com/Chintan2108/SCOR_Workshop_2025)
3. Attempt to install the environment using the “.yaml” files in the repository. These contain the packages needed for the exercises. See steps below:
  - a. Clone the GitHub repository to a location of your choice on your local machine
  - b. Open Anaconda prompt (cmd), and navigate to the repository
  - c. Run command:  
`conda env create -f scor_2025_python_env.yml`
  - d. If you see a “environment created successfully” message, run command:  
`conda activate scor_2025`
  - e. If step “c” was unsuccessful, run command:  
`conda env create -f scor_2025_python_env_cross-plt.yml`  
and repeat step “d” again.

NOTE: Try to set things up on your own following the above instructions. In case you are stuck or unable to proceed further, do not worry, we will spend the first 10-15 minutes ensuring everyone is set up properly and ready to go. Feel free to reach out to me at [chintanmaniyar@uga.edu](mailto:chintanmaniyar@uga.edu) in case of any questions/queries.